

Telemetry-Software Integration

Technical Guidelines

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Abstract

This document defines the telemetry requirements to be implemented by **the Software Division**. It specifies the telemetry signals to be transmitted to the Jetson Nano, their acquisition methods, and formatting guidelines. It also defines the placement and presentation of telemetry data within the rover's GUI, including real-time readings, statistics, trouble codes, warnings, and other system-status information. Preliminary design requirements cover the number and type of signals to be monitored, methods for reading specific signals, GUI functionality, and troubleshooting considerations. The document serves as a reference for implementing the software side of the rover's telemetry system.

Requirements & Objectives

2.1 Data Acquisition

The telemetry serial link shall operate at a **115200 baudrate**. The Microcontroller shall initialize its USB serial interface using [*Serial.begin(115200)*], while the Jetson Nano shall configure the corresponding Linux serial device to the same communication rate.

Telemetry data shall be transmitted through the serial interface using a structured, ASCII-encoded frame with the following format:

XXX=##.##,XXX=##.##\n

Each telemetry variable shall consist of a predefined **variable identifier**, followed by the assignment character (=) and its corresponding numerical value. Individual variables shall be separated by a comma (,), while each complete telemetry frame shall be terminated by a newline character (\n).

For a more detailed look at the specific signals to be read, see **Appendix A**.

2.2 Signal Quality & Accuracy

As previously specified, each telemetry signal shall be represented with a minimum numerical resolution of **two decimal places**. All telemetry values shall be appropriately scaled, converted, and validated for measurement accuracy prior to serial transmission. Signal acquisition, calibration, conversion, and validation shall remain the responsibility of the **Power & Electrical Systems Division**.

Complete telemetry frames shall be transmitted to the Jetson Nano at a fixed rate of **4 frames per second (4 Hz)**, corresponding to a transmission interval of **250 ms per frame**.

2.3 Telemetry Data on GUI

Each telemetry signal shall be individually inspectable through the GUI. For every received signal, the interface shall display its **current value**, **minimum recorded value**, **maximum recorded value**, and **average value since system startup**. Each signal shall additionally have an associated **time-series graph** displaying its historical value as a function of elapsed time, using the appropriate engineering unit (V, A, °C, etc.).

The displayed current value and corresponding time-series graph shall be updated at the specified telemetry frame rate of **4 Hz**, resulting in one GUI data update every **250 ms**. Upon reception and successful parsing of each telemetry frame, the

GUI shall append the new sample to the corresponding signal history and refresh the displayed current, minimum, maximum, and average values.

2.4 Troubleshooting & Diagnostics

The lowercase identifier **f** shall designate a given field as a **fault variable**, while **G** shall represent the **fault severity level** and **X** shall represent a unique fault index used exclusively to distinguish individual fault entries within the telemetry frame.

The four-digit numerical value shall encode the origin and type of the detected fault using a predefined hierarchical structure. The **first digit** shall identify the rover subsystem in which the fault originated. The **second digit** shall identify the specific telemetry signal within that subsystem responsible for generating the fault. The **remaining two digits** shall identify the specific type of fault detected for that signal.

The general fault-code structure shall therefore be defined as:

fXX=SDGF

where the individual digits represent:

- **G** – Fault severity level
- **XX** – Unique fault index within the transmitted telemetry frame
- **S** – Subsystem identifier
- **D** – Signal identifier within the specified subsystem
- **F** – Fault-type identifier

For example, the transmitted field:

f12=3412

shall identify a fault with **fault index 12**, originating from **subsystem 3**, associated with **signal 4** within that subsystem, having **severity level 0**, and corresponding to **fault type 2**.

The Software Division shall decode received fault codes according to the **fXX=SDGF** structure and use the subsystem and signal identifiers to associate each fault with its corresponding telemetry element within the GUI. The **severity identifier (G)** shall determine the predefined severity classification of the detected condition, while the **fault-type identifier (F)** shall be used to retrieve the corresponding fault description.

The **fault index (XX)** shall serve exclusively as a unique identifier for distinguishing individual fault entries within the transmitted telemetry frame and shall not be used to determine the origin, severity, or type of the fault.

For a complete definition of severity levels, subsystem identifiers, signal identifiers, fault-type identifiers, and their corresponding descriptions, refer to **Appendix B**.

2.5 Exporting Telemetry Data

The Software Division shall implement a feature allowing all data collected during the current operating session to be exported as an **.xlsx workbook**. Session telemetry shall be stored continuously in memory or an appropriate local data structure as frames are received, allowing the complete dataset to be written to the workbook upon an export request.

The exported workbook shall contain:

- **Telemetry Data:** All signal readings organized against elapsed time since system startup.
- **Signal Statistics:** Minimum, maximum, and average values for each telemetry signal.
- **Fault Log:** All registered fault events, including fault code, corresponding message, subsystem, originating subsystem signal, and session timestamp.

The Software Division shall be responsible for implementing the .xlsx generation process and determining the workbook structure and formatting. Export operations shall not interrupt the established **4 Hz telemetry acquisition process**.

Preliminary Design Specifics

3.1 Warnings & Faults Displayed

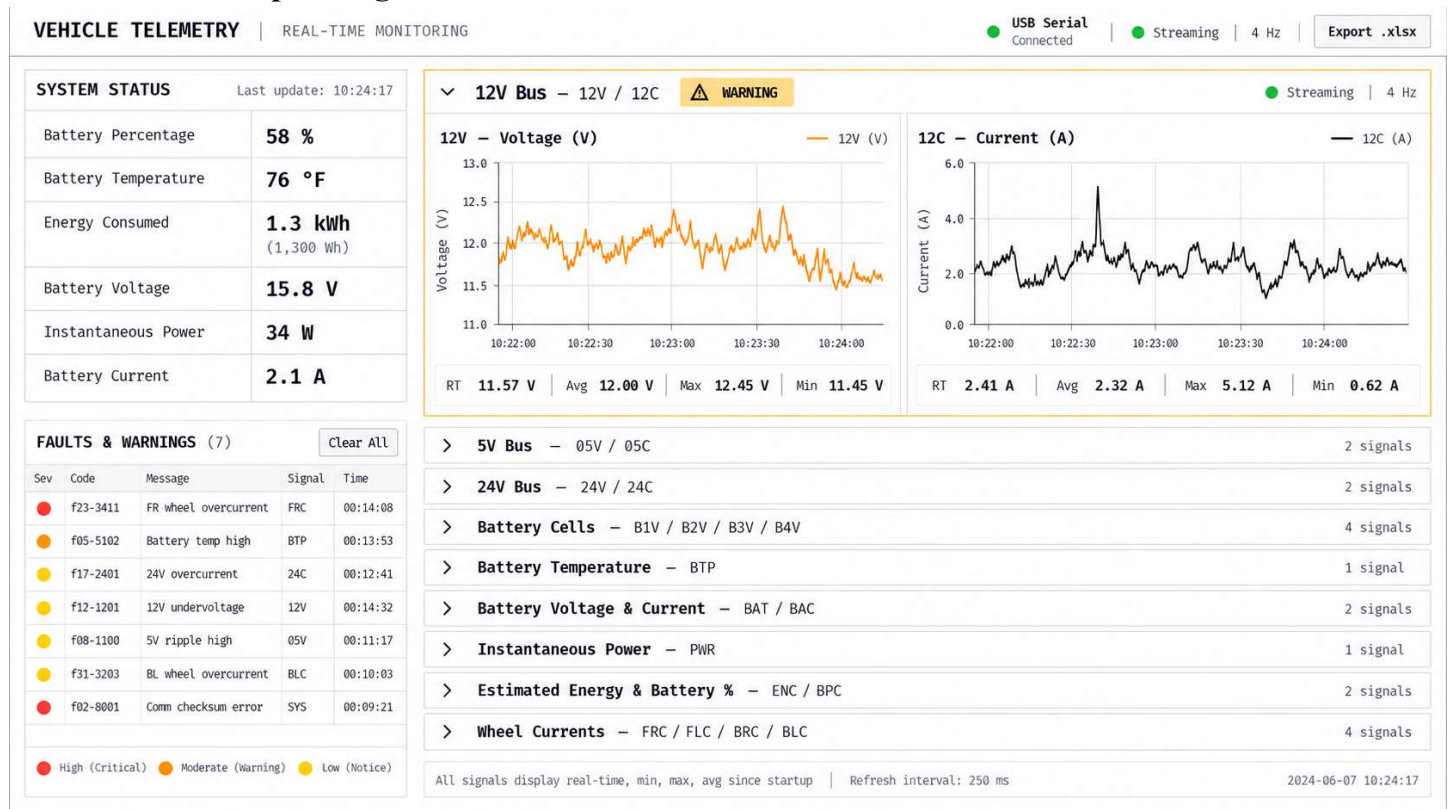
Trouble codes shall remain visible and accessible at all times through the GUI. When a fault is detected, the GUI element associated with the affected telemetry signal shall be highlighted according to its assigned fault severity level:

- **Yellow** – Low severity/warning
- **Orange** – Moderate severity
- **Red** – High severity

The GUI shall include a dedicated Fault/Warning menu containing all recorded faults. Each entry shall display the fault code, corresponding fault message, severity level, affected signal, and timestamp of occurrence. Fault entries shall be prioritized by severity level, with higher-severity faults displayed first. Faults of equal severity shall be ordered by their time of occurrence.

The Software Division shall be responsible for decoding each received fault variable, associating it with the appropriate subsystem and signal, retrieving the corresponding fault message, and applying the appropriate severity-based visual indication within the GUI.

3.2 GUI Concept Design



The previous image shall serve only as a **rough visual concept** for the telemetry GUI and therefor not be considered a final design specification. The **Software Division will retain authority over the final GUI layout, styling, and implementation**, and may modify the proposed design as necessary to ensure that the interface is practical, maintainable, and achievable within the project's software constraints.

Certain functional characteristics of the concept are nevertheless recommended. The **System Status** and **Faults & Warnings** sections should remain **continuously visible and static** so that critical system information can be monitored regardless of which telemetry signals are being viewed. The System Status section should display selected high-level or derived parameters, such as **estimated battery percentage, battery voltage, battery current, battery temperature, instantaneous power, and estimated energy consumed**. The Faults & Warnings section should provide persistent access to active or previously detected fault conditions, including the corresponding fault code, affected signal or subsystem, descriptive message, severity, and timestamp.

Individual telemetry signals should be presented within a **scrollable region separate from the static status panels**. To improve readability and reduce screen clutter, the use of **collapsible/drop-down sections organized by subsystem or signal group** is recommended. Expanding a section should expose the corresponding real-time signal plots and associated statistics, while collapsed sections should remain compact. This arrangement is intended to allow the operator to focus on selected telemetry signals without losing visibility of system-level status or fault information.

Appendix

A. DATA IDENTIFIERS

<i>Identifier</i>	<i>Signal</i>	<i>Unit</i>
12V	12 V Bus Voltage	V
12C	12 V Bus Current	A
05V	5 V Bus Voltage	V
05C	5 V Bus Current	A
24V	24 V Bus Voltage	V
24C	24 V Bus Current	A
BTP	Battery Temperature	°F
B1V	Battery Cell 1 Voltage	V
B2V	Battery Cell 2 Voltage	V
B3V	Battery Cell 3 Voltage	V
B4V	Battery Cell 4 Voltage	V
FRC	Front-Right Wheel Current	A
FLC	Front-Left Wheel Current	A
BRC	Back-Right Wheel Current	A
BLC	Back-Left Wheel Current	A
BAT	Battery Voltage	V
BAC	Battery Current	A
—	DERIVED SIGNALS	—
PWR	Instantaneous Power	W
ENC	Estimated Energy Consumed	Wh
BPC	Estimated Battery Percentage	%

B. FAULT CODES

B.1 Fault Severity Levels

<i>G</i>	<i>Severity Level</i>	<i>Classification</i>
1	Warning	Abnormal condition requiring operator awareness.
2	Critical	Condition requiring prompt operator attention or corrective action.
3	Emergency	Condition requiring immediate operator or system response.

B.2 Subsystem Identifiers

<i>S</i>	<i>Subsystem</i>	<i>D</i>	<i>Signal</i>
1	Power Distribution	1	12 V Bus Voltage
1	Power Distribution	2	12 V Bus Current
1	Power Distribution	3	5 V Bus Voltage
1	Power Distribution	4	5 V Bus Current
1	Power Distribution	5	24 V Bus Voltage
1	Power Distribution	6	24 V Bus Current
2	Battery System	1	Battery Voltage
2	Battery System	2	Battery Current
2	Battery System	3	Battery Temperature
2	Battery System	4	Battery Cell 1 V
2	Battery System	5	Battery Cell 2 V
2	Battery System	6	Battery Cell 3 V
2	Battery System	7	Battery Cell 4 V
2	Battery System	8	Estimated Battery Percentage
2	Battery System	9	Estimated Energy Consumed
3	Mobility System	1	Front-Right Wheel Current
3	Mobility System	2	Front-Left Wheel Current
3	Mobility System	3	Back-Right Wheel Current
3	Mobility System	4	Back-Left Wheel Current
4	Thermal / Environmental Monitoring	1	Primary Temperature Signal

B.3 Fault-type Identifiers

<i>F</i>	<i>Fault Type</i>	<i>Definition</i>
1	High Limit	Signal exceeds its defined maximum threshold.
2	Low Limit	Signal falls below its defined minimum threshold.
3	Invalid Reading	Signal value falls outside its physically or logically valid range.
4	Signal Lost	Expected signal is unavailable.